Composing layers

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

Before you start

For z = -2 and z = 3, what does ReLU return? Does this operation itself assign a class?

Plan time to read, investigate and explain. Keep predictions separate from observations.

Your mission

> Start with the target: Trace how one layer supplies the next layer inputs. Try the recall before opening notes.

Your learning target

I can trace how one **layer** supplies the next **layer input**s.

I can check that connected **layer** dimensions and feature order agree.

I can explain why intermediate activations are not automatically class **prediction**s.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Activation functions. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Dot products and weighted sums. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

This week you will practise:

Trace how one **layer** supplies the next **layer input**s

Check that connected **layer** dimensions and feature order agree

Explain why intermediate activations are not automatically class **prediction**s

Retrieve what you already know

> For z = -2 and z = 3, what does ReLU return? Does this operation itself assign a class?

> **My starting explanation**

This week's end goal

By the end, I can explain composing **layer**s using a traced example and a stated limitation.

STUDENT LESSON · 2 OF 18

Notice the evidence

> Investigate the supplied Composing layers case. Separate observations, given facts and assumptions before choosing an explanation.

A network connects **layer**s so that the **output** vector of one **layer** becomes the **input** vector of the next. Use a fictional numerical **input** x = [x1, x2].

The first **layer** computes z1 = x1 - x2 and z2 = x1 + x2 - 2, then applies ReLU separately: h1 = max(0, z1), h2 = max(0, z2). Its **output** is the ordered vector h = [h1, h2].

The next **layer** computes y = 2h1 - h2 + 1. It receives h, not the original x. It uses weights [2, -1] and bias 1, with identity as its **output** activation.

This is a two-**layer** computation when counting the **layer**s with weights; an **input** placeholder is not counted here. All parameters are supplied, so this lesson performs no **training**.

Stage	Input size	Output size
First layer with ReLU	2 components	2 components
Output layer with identity	2 components	1 component

The two first-**layer** units both receive the same original x. They can be calculated in either order. Only after both activations are available do we evaluate the **output layer**.

> **List the given facts and one fact that is not supplied.**

STUDENT LESSON · 3 OF 18

See how it works

> Follow the mechanism for Composing layers. Connect each labelled input, deciding step and result.

Take x = [3, 1]. First compute z1 = 3 - 1 = 2 and z2 = 3 + 1 - 2 = 2. ReLU gives h = [2, 2]. The **output layer** receives exactly this ordered pair and returns y = 2 times 2 - 2 + 1 = 3.

Record z, h and y separately so that a correct final number cannot hide an incorrect intermediate step.

```
def forward(x):
    if len(x) != 2:
        raise ValueError("Expected two input components")
    x1, x2 = x
    z = [x1 - x2, x1 + x2 - 2]
    h = [max(0, value) for value in z]
    y = 2 * h[0] - h[1] + 1
    return z, h, y

print(forward([3, 1]))  # ([2, 2], [2, 2], 3)
print(forward([1, 3]))  # ([-2, 2], [0, 2], -1)
```

The code assumes numeric **input**s with the agreed meaning. Its length check does not validate units or field order. Python indices 0 and 1 correspond to the first and second hidden components.

There is no feedback from y into h during this **forward calculation**.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

STUDENT LESSON · 4 OF 18

Compare the cases

> Compare the supplied cases for Composing layers. Identify the detail that changes the conclusion.

Compare a familiar case with a changed case

A revised hidden **layer** returns [h1, h2, h3]. The **output layer** still has weights [2, -1]. Explain the mismatch and what must be specified before calculation.

Keep the procedure fixed. Explain what changed in the inputs and result.

Change one thing in the pictured case

Change one thing

Change: Change only v from 3 to1.

Predicted result

Prediction: h1 stays 2, h2 becomes 0, y becomes 2.

Why the result follows

Reason: The second hidden unit’s preactivation reaches 0 while the first unit is unchanged.

STUDENT LESSON · 5 OF 18

Words that unlock the idea

Use the vocabulary of Composing layers to explain a concrete decision, rather than only memorising definitions.

Word	Meaning
Layer	A group of computations that maps an input vector to an output vector

Word	Meaning
Hidden activation	An intermediate value computed inside a network
Composition	Applying one function to the output of another
Forward calculation	Computing successive values from an input using fixed parameters

Use two terms to explain a step in this week's example.

STUDENT LESSON · 6 OF 18

Follow a worked example

Follow the worked trace for Composing layers, predict a step, then compare with its explanation.

Complete each stage before using its output

AVAILABLE INPUT	COMPUTATION	VALUE PASSED ON
x is [3, 1]	Both first layer scores	z is [2, 2]
z is [2, 2]	ReLU on each component	h is [2, 2]
h is [2, 2]	2h1 - h2 + 1	y is 3

Read the rows from top to bottom. The first two rows together form the hidden layer; the final row is the output layer. Each row shows the values actually passed to the next stage.

Trace x = [1, 1]. Record z1, z2, h1, h2 and y. Explain why using x directly in the output formula would be a different model.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Composing layers

Compose two hidden detectors before one **output** instead of skipping a **layer**.

The facts we will use

Supplied facts

For **input** [u, v], h1=ReLU(u-2), h2=ReLU(v-1), y=h1-h2. **Input** [4, 3].

Supplied facts

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-5/5

Composing layers

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. First hidden unit

u-2=2, so h1=2.



2. Second hidden unit

v-1=2, so h2=2.



3. Compose output

y=2-2=0; the output uses activated hidden values, not raw inputs.

CHANGE ONE THING

Change only v from 3 to1.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. First hidden unit	u-2=2, so h1=2.
2. Second hidden unit	v-1=2, so h2=2.
3. Compose output	y=2-2=0; the output uses activated hidden values, not raw inputs.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Composition means one **layer**'s result becomes the next **layer**'s **input**. Compute both hidden units before combining them. At [4,3], their equal **output**s cancel.

Lowering only the second **input** removes its positive hidden contribution, so subtracting it no longer reduces the **output**. The final subtraction must use activated hidden values rather than the raw measurements.

What this example does not establish

Keep the stated limit

Hand-composed detectors illustrate **layer** order; they are not **evidence** of a learned or biologically realistic neural **mechanism**.

Keep the stated limit

Read the labelled picture: What intermediate value explains the changed **output**? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Keep the stated limit

Change one thing: Change only v from 3 to1. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Composing layers claim and explain a repair using the supplied evidence.

Check the claim against the facts

The **output layer** expects two **input**s because its two weights match h1 and h2. If the hidden **layer** is redesigned to return three activations, the **output layer** needs a compatible three-**input** definition.

Silently dropping an activation or inventing a third weight does not preserve the stated model.

Check the claim against the facts

Order matters even when lengths match. For x = [4, 1], h = [3, 3] happens to be unchanged by swapping, so that case cannot reveal an order bug. For x = [3, 0], h = [3, 1] gives y = 6.

Swapping only those hidden values gives 2 times 1 - 3 + 1 = 0. Swapping both the values and their matching weights would preserve the dot product.

Keeping the **output layer** fixed, replacing ReLU with identity changes the model. For x = [1, 3], identity gives h = [-2, 2] and y = -5; ReLU gives h = [0, 2] and y = -1.

More **layer**s or a different activation do not establish better **prediction**s without valid evaluation.

Locate the earliest unsupported step

Compare hidden vectors for x = [4, 1] and x = [3, 0]. Which **input** reveals a bug that swaps h1 and h2 but keeps **output** weights unchanged? Give correct and swapped **output**s for that **input**.

State one repair and the checks you would rerun.

STUDENT LESSON · 8 OF 18

Practise together

Practise Composing layers with support, then explain the deciding step in your own words.

For x = [1, 3], write both scores, both ReLU activations and y using the lesson model.

Trace x = [1, 1]. Record z1, z2, h1, h2 and y. Explain why using x directly in the output formula would be a different model.

Explain your prediction to a partner before comparing results.

Practise with less help: Repair a dimension mismatch

Use workbook W2 for the connected example “Composing **layer**s”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Composing **layer**s. Proposed explanation: “The second hidden unit receives the first hidden unit **output**”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

What intermediate value explains the changed **output**? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

Run the investigation

> Investigate Composing layers. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

Predict, investigate and preserve the evidence

A revised hidden **layer** returns [h1, h2, h3]. The **output layer** still has weights [2, -1]. Explain the mismatch and what must be specified before calculation.

```
def forward(x):
    if len(x) != 2:
        raise ValueError("Expected two input components")
    x1, x2 = x
    z = [x1 - x2, x1 + x2 - 2]
    h = [max(0, value) for value in z]
    y = 2 * h[0] - h[1] + 1
    return z, h, y

print(forward([3, 1]))  # ([2, 2], [2, 2], 3)
print(forward([1, 3]))  # ([-2, 2], [0, 2], -1)
```

Record	Your evidence
Given input and fixed rule	
Prediction before checking	
Actual result, or “not run”	
First mismatch and one explanation	

Record	Your evidence
Boundary or missing input and expected response	

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

STUDENT LESSON · 10 OF 18

Build your understanding

> Move from recall to tracing, diagnosis and transfer in Composing layers. Explain why each step is justified.

Explain → apply → challenge

For x = [1, 3], write both scores, both ReLU activations and y using the lesson model.

Compare hidden vectors for x = [4, 1] and x = [3, 0]. Which **input** reveals a bug that swaps h1 and h2 but keeps **output** weights unchanged? Give correct and swapped **output**s for that **input**.

> Use the lesson model to verify h and y for [0, 1] and [0, 2]. Could a fixed later **layer** receiving only h distinguish the two original **input**s?

Explain which evidence supports your conclusion and which claim remains unproved.

STUDENT LESSON · 11 OF 18

Connect your project

> Apply this week's idea to Image-pattern Network Investigation. Keep an annotated decision and the evidence supporting it.

Add a labelled composing **layer**s **evidence** sheet to Image-pattern Network Investigation. Show how you can trace how one **layer** supplies the next **layer input**s.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Project evidence	Record
Component and version	
Inputs permitted at prediction time	
Trace and comparison with earlier work	
Failure, limit and human review	

Evidence for your project

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

STUDENT LESSON · 12 OF 18

Show what you know

> Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

The computed y is a numerical **output**, not a verified probability or a class by itself. If we separately declare class 1 when y >= 0 and class 0 otherwise, the worked **input**s [3, 1] and [1, 3] receive classes 1 and 0.

Their true labels are not supplied, so these calculations do not measure accuracy.

A hidden value is a computed feature. We should not name h1 as confidence or h2 as danger without **evidence** for that interpretation.

Distinct **input**s may also produce the same hidden vector: [0, 1] and [0, 2] both give h = [0, 0] and y = 1. A later deterministic **layer** seeing only h cannot recover the lost distinction.

> **A team says its output is correct because all layers connect and its code runs. Which claim has it checked, and what evidence is still missing?**

Next week uses longer forward-pass tables to locate the first incorrect intermediate value and check complete network calculations.

> Use the lesson model to verify h and y for [0, 1] and [0, 2]. Could a fixed later **layer** receiving only h distinguish the two original **input**s?

My limited claim and a useful next check

STUDENT LESSON · 13 OF 18

Study a complete case

Revisit the complete Composing layers case for supported practice. Repeated examples do not become fresh assessments.

Complete evidence record

Complete each stage before using its output

AVAILABLE INPUT	COMPUTATION	VALUE PASSED ON
x is [3, 1]	Both first layer scores	z is [2, 2]
z is [2, 2]	ReLU on each component	h is [2, 2]
h is [2, 2]	2h1 - h2 + 1	y is 3

Read the rows from top to bottom. The first two rows together form the hidden layer; the final row is the output layer. Each row shows the values actually passed to the next stage.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

STUDENT LESSON · 14 OF 18

Try a new case independently

Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Compose a new network independently

A new model takes [u, v]. Compute a = ReLU(u + v - 3), b = ReLU(u - v), then q = a + 2b - 1. Trace both **input**s [2, 1] and [1, 2], including scores before ReLU. Apply class 1 when q >= 0.

State the hidden and **output** vector sizes.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Attempt this without hints

A new model takes [u, v]. Compute a = ReLU(u + v - 3), b = ReLU(u - v), then q = a + 2b - 1. Trace both **input**s [2, 1] and [1, 2], including scores before ReLU. Apply class 1 when q >= 0.

State the hidden and **output** vector sizes.

> **My independent procedure, result and limitation**
>
> | |
> |---|
> | |
> | |
> | |
> | |

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

Your lesson route

> Use this map to navigate Composing layers, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: Compose two hidden detectors before one **output** instead of skipping a **layer**.

By the end, I can	How I will show it
Trace how one layer supplies the next layer inputs	Explain the deciding step, show a trace or test, and state one limit.
Check that connected layer dimensions and feature order agree	Explain the deciding step, show a trace or test, and state one limit.
Explain why intermediate activations are not automatically class predictions	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Composing layers without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Trace how one **layer** supplies the next **layer input**s

> **My deciding step, evidence and one remaining question**

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Compose two hidden detectors before one **output** instead of skipping a **layer**.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? y=2-2=0; the **output** uses activated hidden values, not raw **input**s.

1. Compose **output**

2. Second hidden unit

3. First hidden unit

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Compose **output** y=2-2=0; the **output** uses activated hidden values, not raw **input**s. **Composition** means one **layer**'s result becomes the next **layer**'s **input**.

Compute both hidden units before combining them. At [4,3], their equal **output**s cancel. Lowering only the second **input** removes its positive hidden contribution, so subtracting it no longer reduces the **output**.

The final subtraction must use activated hidden values rather than the raw measurements.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Change only v from 3 to1. Which conclusion is supported by the supplied facts?

1. h1 stays 2, h2 becomes 0, y becomes 2.

2. We can decide the result without examining the changed **input** or the procedure.

3. This one example guarantees correct results for every future **input**.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: h1 stays 2, h2 becomes 0, y becomes 2. The second hidden unit's preactivation reaches 0 while the first unit is unchanged.

Repair a missing building block

Review Activation functions (ai-5/4). Goal: Trace identity ReLU and sigmoid on a numerical score. Short practice: Which original **input**s cannot be distinguished by this network's final **output**?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Dot products and weighted sums (ai-5/2). Goal: Calculate a dot product by matching feature positions. Short practice: Which connection explains why more right brightness lowers this score?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

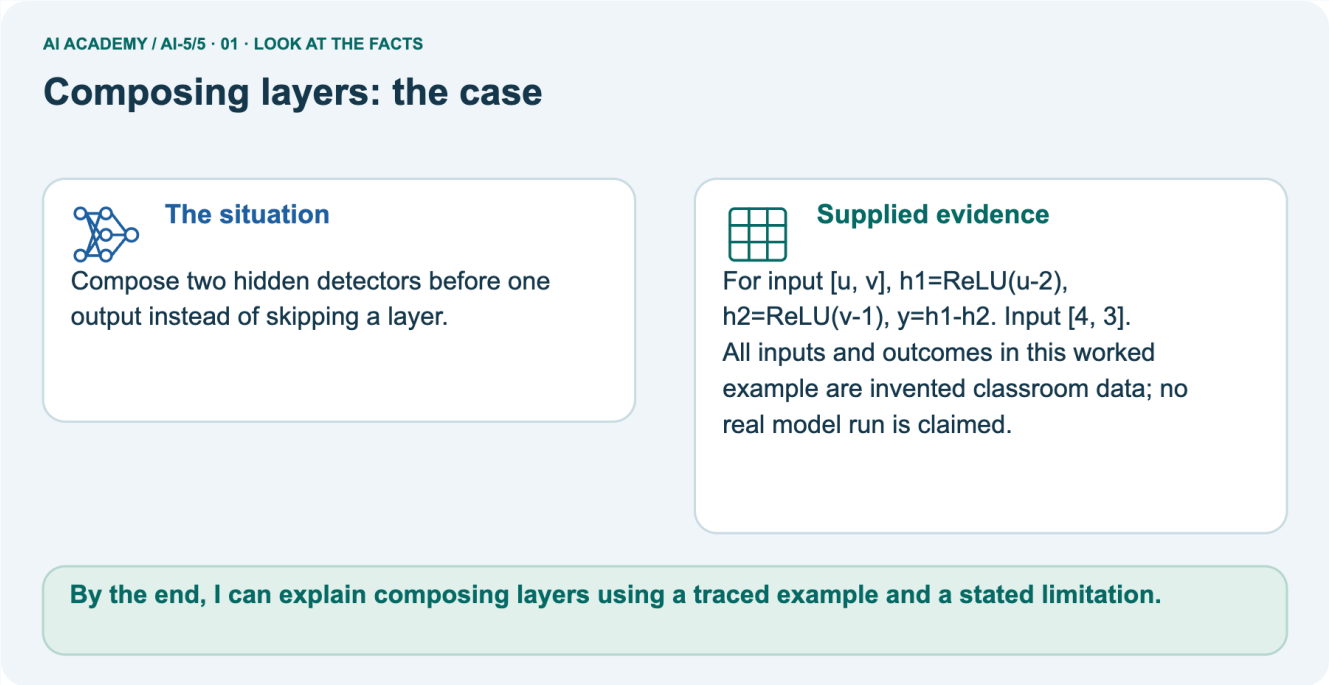
Drafts stay in this browser when storage is available; clear drafts on a shared device.

STUDENT LESSON · 18 OF 18

Visual concept journey

> Explain each Composing layers graphic using its labels, deciding evidence and relevant limitation.

01. Composing layers: the case



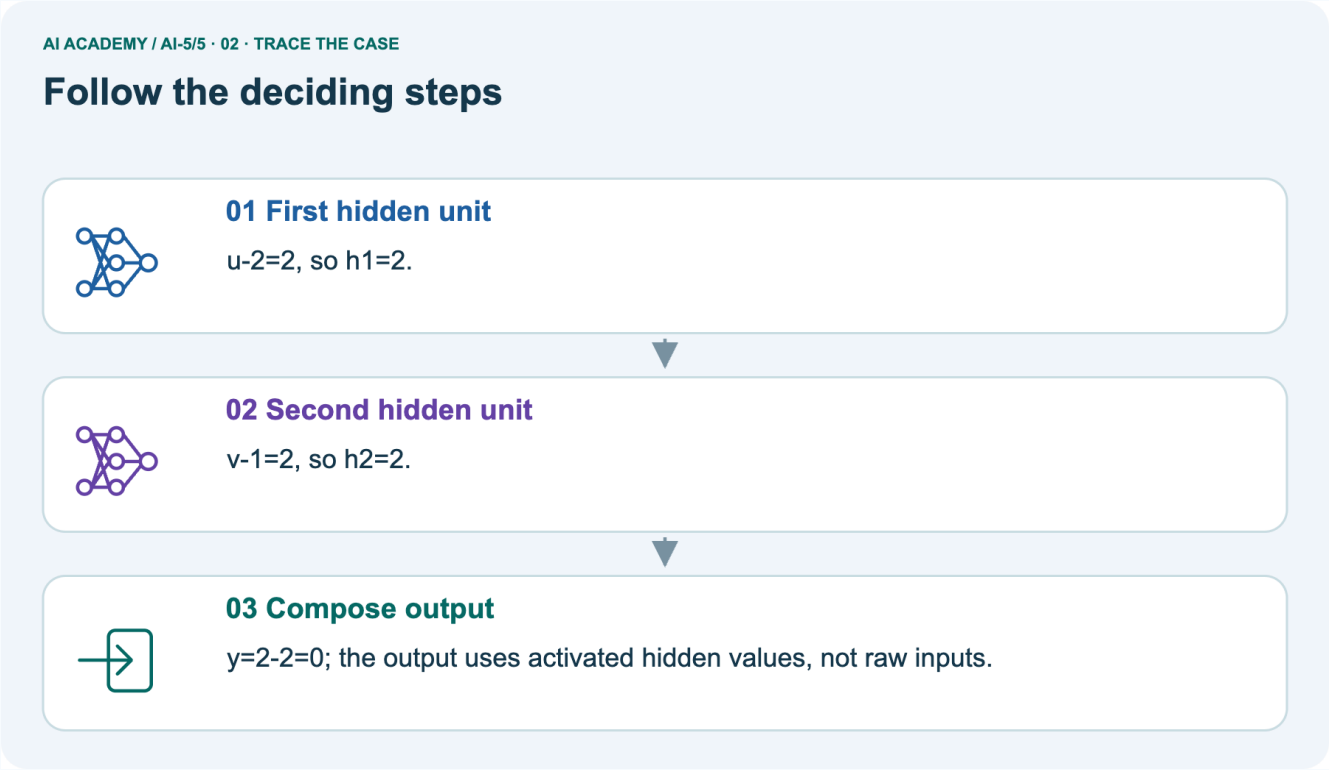
Follow the labelled picture with the matching explanation. Open the vector diagram

Compose two hidden detectors before one **output** instead of skipping a **layer**. Supplied facts: For **input** [u, v], h1=ReLU(u-2), h2=ReLU(v-1), y=h1-h2. **Input** [4, 3].

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

> Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

First hidden unit: u-2=2, so h1=2. → Second hidden unit: v-1=2, so h2=2. → Compose **output**: y=2-2=0; the **output** uses activated hidden values, not raw **input**s.

> Point to each numbered stage and explain which recorded **input** it uses.

03. Complete each stage before using its output

AI ACADEMY / AI-5/5 · 03 · INSPECT THE LESSON MODEL

Complete each stage before using its output

Available input	Computation	Value passed on
x is [3, 1]	Both first layer scores	z is [2, 2]
z is [2, 2]	ReLU on each component	h is [2, 2]
h is [2, 2]	2h1 - h2 + 1	y is 3

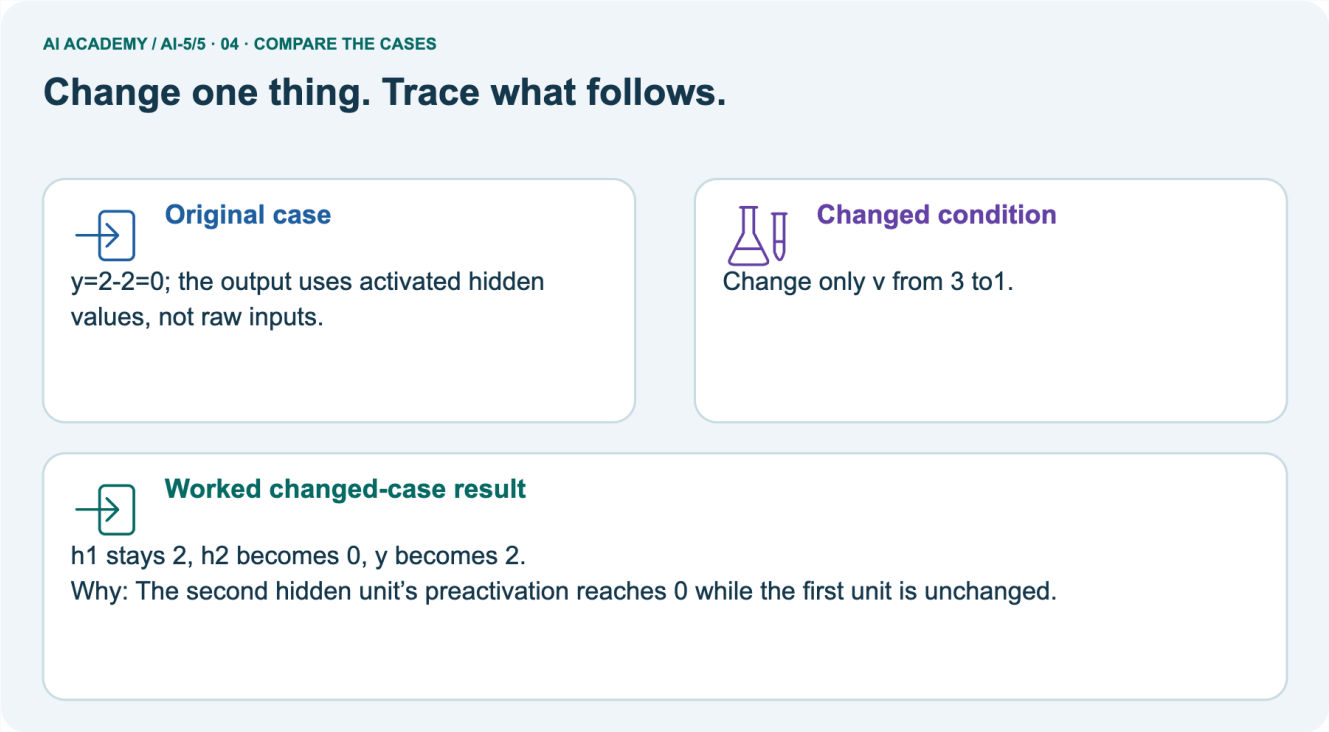
Read the rows from top to bottom. The first two rows together form the hidden layer; the final row is the output layer. Each row shows the values actually passed to the next stage.

Follow the labelled picture with the matching explanation. Open the vector diagram

Complete each stage before using its **output**. x is [3, 1] | Both first **layer** scores | z is [2, 2]. z is [2, 2] | ReLU on each component | h is [2, 2]. h is [2, 2] | 2h1 - h2 + 1 | y is 3

> Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



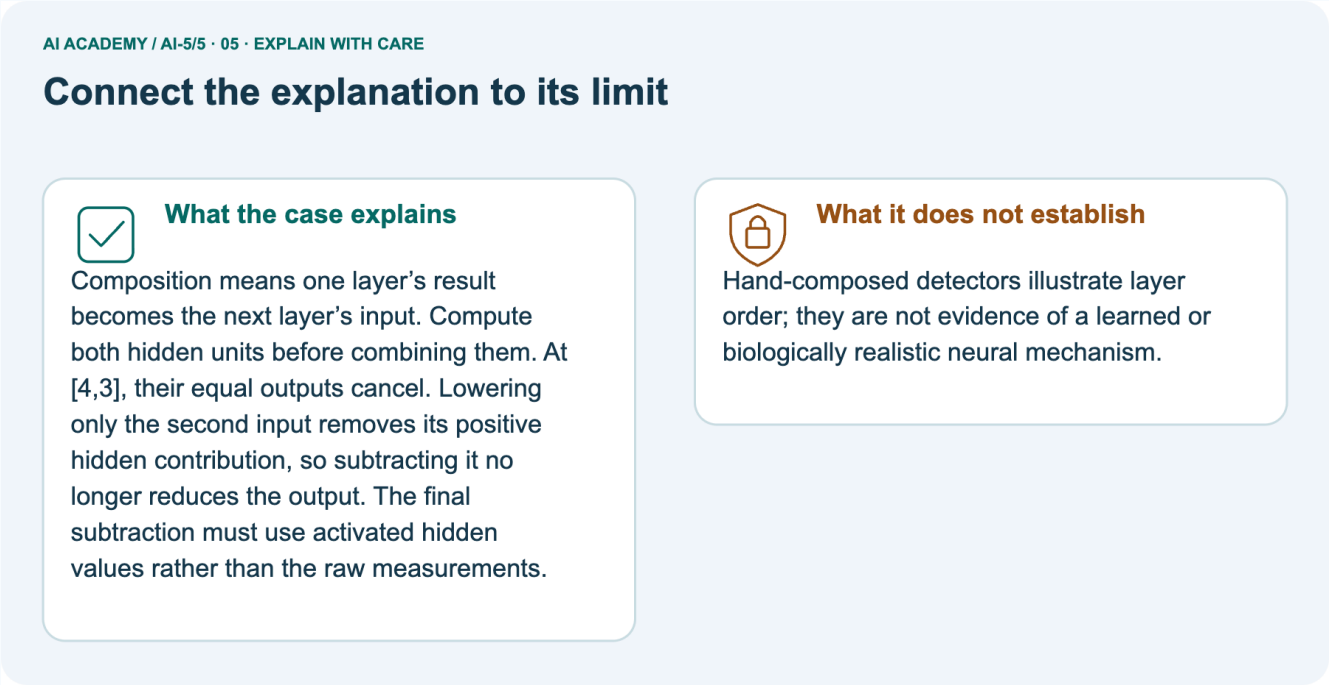
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: y=2-2=0; the **output** uses activated hidden values, not raw **input**s. Changed condition: Change only v from 3 to1. Expected result: h1 stays 2, h2 becomes 0, y becomes 2.

Reason: The second hidden unit's preactivation reaches 0 while the first unit is unchanged.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

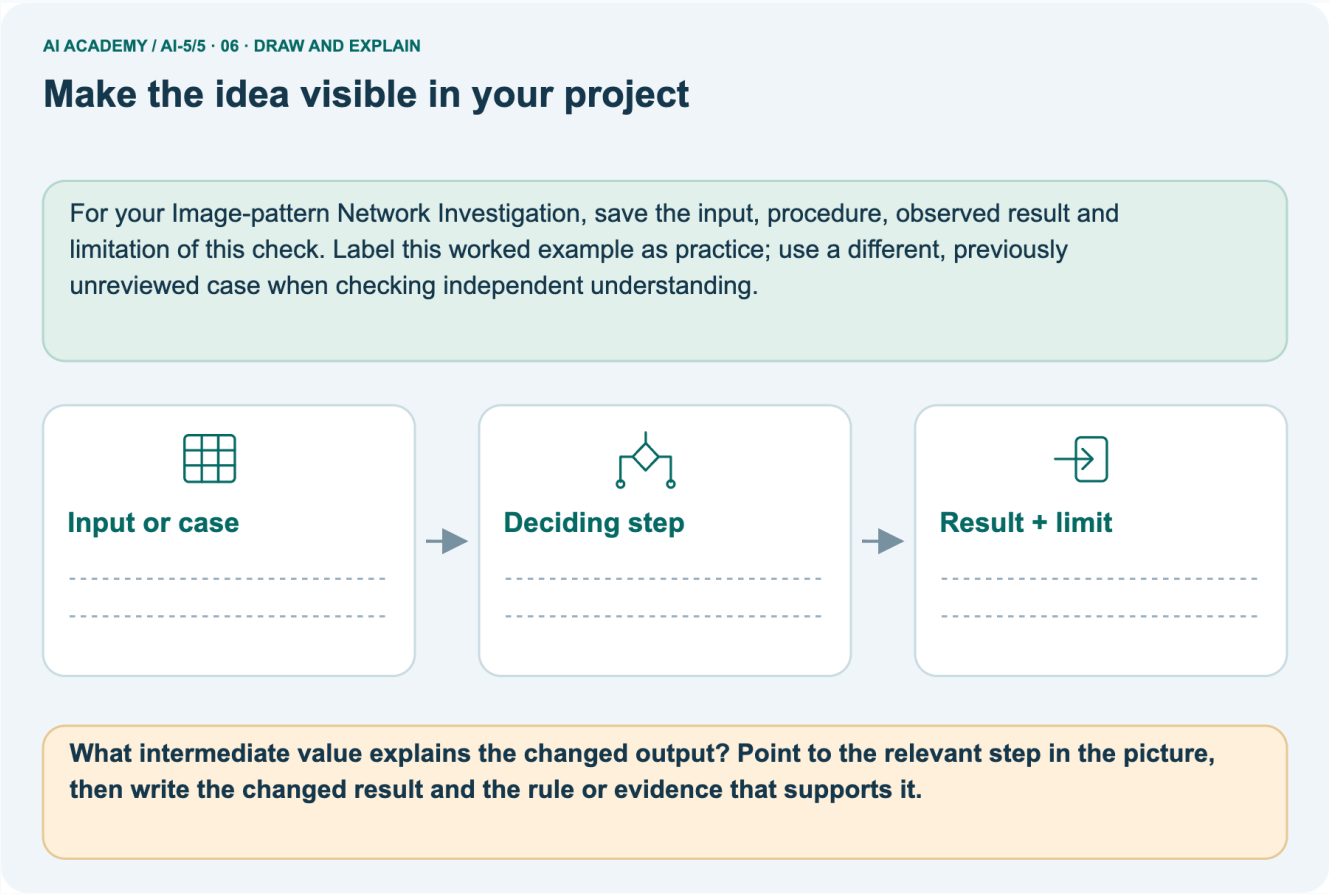
Composition means one **layer**'s result becomes the next **layer**'s **input**. Compute both hidden units before combining them. At [4,3], their equal **output**s cancel.

Lowering only the second **input** removes its positive hidden contribution, so subtracting it no longer reduces the **output**. The final subtraction must use activated hidden values rather than the raw measurements.

Limit: Hand-composed detectors illustrate **layer** order; they are not **evidence** of a learned or biologically realistic neural **mechanism**.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: What intermediate value explains the changed **output**? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.